Course Workshop Plugins — Planning Pipeline 执行全记录

> 执行时间：2026-03-28 执行环境：astra-studio-plugins (作为插件开发平台) 目标项目：courses-workshop-plugins 输入材料：IM-PreK.3-Nov果汁-PBL.pdf + 三维项目阶段图-en.xlsx

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0. 前置准备：输入材料分析

0.1 读取 PDF — PBL 项目活动预案

```
Read /Users/liuyameng/Desktop/好奇贝果/PBL 课程设计/IM-PreK.3-Nov果汁-PBL.pdf
  → PDF pages extracted: 20 page(s) (9.3MB)
```

提取到的预案结构（华美 PBL 项目路径图 5 段式）：

- 01 项目概览 Project Overview — 主题背景 + 教育价值阐述
- 02 项目标准 Project Standard — 先前经验 + 4C能力映射 + 学习目标
- 03 项目启动 Project Launch — 驱动性问题 + 主题网络图
- 04 项目探究 Project Inquiry — 3条递进探究线索，每条含多个活动
- 05 项目展示 Project Products — 成果展示方案

提取到的活动编码规范： `PBL-C{线索号}-{序号}`

案例主题： "我周围的人 / People Around Me" **驱动性问题：** "如何开一家果汁店？ / How to open a juice shop?"

3 条探究线索： – Clue 1: 果汁店要做什么口味的果汁？(3–5 days) – Clue 2: 如何经营果汁店？(3–5 days) – Clue 3: 开业前需要做哪些准备？(3–5 days)

活动详情提取（每条线索 4–9 个活动）：

Clue 1 Activities: | Code | Name | Key Question | |------|------|-------------| | PBL-C1-01 | 我喜欢的果汁 / My Favorite Juice | 要做什么口味的果汁？ | | PBL-C1-02 | 你喜欢什么果汁？ / What Juice Do You Like? | | | PBL-C1-03 | 应季的水果 / Fruits In Season | | | PBL-C1-04 | 营养搭配 / Nutritional Matching | |

Clue 2 Activities: | Code | Name | Key Question | |------|------|-------------| | PBL-C2-01 | 果汁制作工具 / Juice Making Tools | 如何制作果汁？ | | PBL-C2-02 | 果汁制作步骤 / Juice Making Steps | | | PBL-C2-03 | 果汁店海报 / Juice Shop Poster | 如何售卖果汁？ | | PBL-C2-04 | 优惠券 / Coupons | | | PBL-C2-05 | 热果汁 / Hot Fruit Juice | | | PBL-C2-06 | 自制保温瓶 / Homemade Thermos Bottle |

| | PBL-C2-07 | 市场调研 / Market Research | 如何让果汁店赚钱？ | | PBL-C2-08 | 计算利润 / Calculating Profits | | | PBL-C2-09 | 果汁定价 / Juice Price | |

0.2 读取 Excel — 三维项目阶段图

```
openpyxl.load_workbook('三维项目阶段图-en.xlsx')
Sheet1 Dimensions: A1:I20
```

提取到的三阶段九要素框架：

阶段	旅行类比	项目课程阶段	案例
项目开始	1.1 目的需求	2.1 孩子发展的需要(目标)	3.1 引发讨论
	1.2 计划+网络图	2.2 列出网络主题图	3.2 网络图
	1.3 验证(想象)	2.3 小组活动、经验	3.3 提出概念
	1.4 团讨确定	2.4 师生共绘网络图	3.4 已知和未知
项目发展	1.5 专家采访	2.5 专家、实地参观	3.5 兽医图书馆
	1.6 实际开展	2.6 扮演、绘画、书写	3.6 建构兽医诊所
	1.7 调整计划	2.7 重新检视、调整	3.7 检查调整
项目结束	1.8 攻略展示	2.8 报告、展示活动	3.8 记录书展示
	1.9 总结反思	2.9 评估、反思	3.9 评估反思

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1. 逆向分析与插件架构提案

1.1 从产出物逆向推导课研工作流程

```
月度主题确定 → 主题背景撰写 → 儿童先前经验评估
      ↓
4C能力映射 → 学习目标制定
      ↓
驱动性问题设计 → 主题网络图构建
      ↓
探究线索拆分(3条递进) → 每条线索的活动设计
      ↓                      ↓
每个活动: 关键问题 + 内容步骤 + 资源清单 + 教师提示
      ↓
项目展示方案 → 评估标准 → 预案定稿
```

1.2 建议的插件架构

```
course-workshop-plugins/
├── workshop-core/          # 课研工作台（初始化、状态管理）
```

```
├── workshop-insight/        # 课程分析工具包
├── workshop-designer/       # 课程设计流水线
├── workshop-quality/        # 质量保障
└── workshop-resource/       # 资源管理
```

1.3 核心设计流水线映射

```
theme-analysis → prior-knowledge + competency-mapping + driving-question
       ↓
inquiry-scaffold → 拆分出 3 条探究线索
       ↓
activity-design × 3 → 每条线索下的活动序列
       ↓
resource-planner → 资源清单匹配
       ↓
proposal-generate → 完整 PBL 预案文档
```

1.4 与 astra-studio 的对应关系

astra-studio 概念	course-workshop 对应
领域(domain)	月度主题（如"我周围的人"）
事件风暴	主题分析 + 儿童兴趣点发散
Persona	课研主任 / 一线教师 / 园长
Journey Map	课研主任"从选题到定稿"的全流程
Process Flow	三阶段九要素的标准化流程
Hotspot	活动设计耗时最长、资源匹配最易出错
Skill Map	15 个 skills
SKILL.md 骨架	每个 skill 的输入/输出/工作流定义

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2. 工作区初始化

2.1 /studio-core:init 执行（此前已完成）

```
Studio initialized at studio/

  studio/config.yaml    - configuration
  studio/changes/       - active plugin development
  studio/agents/        - custom domain expert definitions
  studio/archive/       - shipped plugin records
```

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3. 阶段一：Event Storm（发现阶段）

3.1 创建领域专家

3 个领域专家写入 `studio/agents/`：

```
Write studio/agents/early-childhood-curriculum-expert.md
  → File created successfully

Write studio/agents/child-development-psychologist.md
  → File created successfully

Write studio/agents/instructional-designer.md
  → File created successfully
```

专家角色说明：

角色	为什么需要	带来的独特视角
产品经理	确保从课研主任的实际痛点出发	用户工作流分析、优先级排序
架构师	评估插件拆分和数据流可行性	技能边界、产出物依赖关系
幼儿教育课程专家	确保 PBL 方法论正确性	三阶段九要素、课标对齐、年龄适配
儿童发展心理学家	确保活动设计符合发展规律	4C能力映射准确性、先前经验评估
教学设计师	确保预案可落地执行	活动序列编排、资源清单完整性、教师可操作性

3.2 领域描述（Step 1: Set the Stage）

> **幼儿园 PBL 课程研发** — 帮助课研主任系统化地设计月度 PBL 项目活动预案。目前课研主任凭经验手工编写，从选题、标准制定、驱动性问题设计、探究线索拆分到活动编排全流程依赖个人能力，缺乏结构化工具支撑，产出质量参差不齐。

执行模式选择： ⚡ **快速模式**（3 个确认点）

3.3 创建 Domain Workspace

```
mkdir -p studio/changes/course-workshop/{personas,journeys,processes}

Write studio/changes/course-workshop/status.json
  → { "type": "domain", "domain": "course-workshop", "phase": "planning",
"plugins": [] }
```

3.4 Step 2–6 合并执行（快速模式）

Persona 1: 张主任（课研主任）

```
Write studio/changes/course-workshop/personas/curriculum-director.md
  → File created successfully
```

核心信息：

```
┌──────────────────────────────────────────────────┐
|  张主任                                           |
|  "我希望有一套标准化的流程，让每个课研老师都能       |
|   产出质量一致的预案"                              |
├──────────────────────────────────────────────────┤
|  Role:         课研主任 / 教研组长                 |
|  Age range:    30-45                              |
|  Tech level:   中                                 |
|                                                   |
|  Pain points:                                     |
|  1. 选题到定稿全流程无工具支撑 - severity: 高       |
|  2. 驱动性问题设计耗时最长，反复修改 3-5 轮 - 高    |
|  3. 4C 能力映射凭感觉 - severity: 中               |
|  4. 资源清单经常遗漏 - severity: 高                |
|  5. 新课研老师上手慢 - severity: 中                |
|                                                   |
|  Success looks like:                              |
|  "30分钟内拿到结构完整的预案草稿，只需做最终审核"   |
└──────────────────────────────────────────────────┘

Pain: 质量依赖个人经验，无法规模化复制
Gain: 一套标准化流程 + 工具，让任何课研老师都能产出合格预案
```

Persona 2: 王老师（一线带班教师）

```
Write studio/changes/course-workshop/personas/classroom-teacher.md
  → File created successfully
```

核心痛点： "预案写得很好看，但到了教室里我不知道第一句话该说什么。"

Persona 3: 李园长

```
Write studio/changes/course-workshop/personas/principal.md
  → File created successfully
```

核心痛点： "我要在 10 分钟内判断一份预案是否合格"

Journey Map: 张主任的月度预案研发流程

```
Write studio/changes/course-workshop/journeys/curriculum-director-monthly-
proposal.md
  → File created successfully
```

6 阶段旅程：

```
┌──────────┬──────────┬──────────┬──────────┬──────────┬──────────┐
| Stage    | 选题研究  | 标准制定  | 问题设计  | 活动编排  | 资源匹配  | 审核定稿  |
├──────────┼──────────┼──────────┼──────────┼──────────┼──────────┤
| Emotion  | 😐 平静  | 😕 困惑  | 😩 焦虑  | 😤 疲惫  | 😕 繁琐  | 😌 释然  |
└──────────┴──────────┴──────────┴──────────┴──────────┴──────────┘
```

Pain Point × Opportunity (8 个机会点):

#	Stage	Pain Point	Severity	Opportunity
1	选题研究	每月重复查找课标	中	主题知识库 + 往年预案索引
2	标准制定	4C 映射无标准化参照	中	自动 4C 映射 + 学习目标生成
3	问题设计	驱动性问题设计是最大瓶颈	高	AI 辅助驱动问题生成 + 开放性验证
4	问题设计	网络图构建耗时	高	自动网络图生成 + 覆盖度检查
5	活动编排	活动设计占整体 70% 时间	高	活动序列自动编排 + 模板化
6	活动编排	探究线索递进性难以把握	中	递进性检查 + 线索平衡分析
7	资源匹配	资源清单遗漏	高	资源自动匹配 + 完整性校验
8	审核定稿	评审反馈碎片化	中	自动质量检查报告

Process Flow 1: 月度 PBL 预案研发流程

```
Write studio/changes/course-workshop/processes/monthly-proposal-creation.md
  → File created successfully
```

5 个决策点:

ID	Decision	Condition
D1	驱动问题是否足够开放且可探究?	问题允许多种答案 + 儿童可动手探索
D2	3 条线索是否递进且均衡?	每条 3–5 天 + 逻辑递进
D3	资源清单是否完整且可获取?	每个活动都有资源 + PBL Box 可供应
D4	评审是否通过?	课标对齐 + 活动可执行 + 资源齐全
D5	园长是否批准?	整体质量合格 + 格式规范

Process Flow 2: 单个探究线索的活动设计流程

```
Write studio/changes/course-workshop/processes/activity-design.md
  → File created successfully
```

3 个决策点（嵌套循环）: – D1: 活动时长 20–30 min? – D2: 活动间逻辑递进? – D3: 覆盖学习目标?

Event Storm 总结 + Hotspot 排名

```
Write studio/changes/course-workshop/event-storm.md
  → File created successfully
```

25 个业务事件，分 6 个事件簇：

事件簇	事件数	关键事件
选题与规划	3	E1 月度主题已确定, E2 课标已对齐, E3 概览已撰写
标准制定	3	E4 先前经验已评估, E5 4C已映射, E6 目标已制定
问题与网络图	3	E7 驱动问题已生成, E8 开放性已验证, E9 网络图已构建
探究线索与活动	6	E10–E15 线索拆分到活动编写
资源与交付	5	E16–E20 资源匹配到文档组装
质量保障	5	E21–E25 检查到审批发放

Hotspot 排名（8 个热点）：

排名	热点	严重度	类型	判断依据
1	HS-1 驱动性问题设计是最大瓶颈	高	知识	Persona 痛点 #2 + Journey 情绪最低点 + Process D1 循环
2	HS-2 活动设计占 70% 时间	高	效率	Journey "活动编排" + Process 嵌套循环
3	HS-3 资源清单遗漏导致开课延误	高	准确	Persona 痛点 #4 + PBL Box 供应链约束
4	HS-4 4C 映射无标准化参照	中	知识	Persona 痛点 #3
5	HS-5 新课研老师上手慢	中	效率	Persona 痛点 #5
6	HS-6 双语内容工作量大	中	效率	Process 独立工作流
7	HS-7 课标对齐靠人工逐条比对	中	准确	课程专家指出
8	HS-8 预案格式不统一	低	效率	园长痛点 #3

专家修正亮点：

- **课程专家**：递进不是"简单→复杂"，而是"具象体验→抽象理解"
- **心理学家**：PDF 中的 4C 映射有误 — PBL-C1-01 标注了 Critical Thinking，实际发展的是 Creativity
- **教学设计师**：PBL-C2-05（保温实验）实际需 40+ 分钟，应拆为 2 课时

3.5 阶段一产出文件清单

```
studio/changes/course-workshop/
├── event-storm.md                              ← 25 个事件 + 8 个热点 + 专家修正
├── status.json                                 ← type: domain, phase: planning
├── personas/
|   ├── curriculum-director.md                  ← 张主任（核心用户）
|   ├── classroom-teacher.md                    ← 王老师（预案消费者）
|   └── principal.md                            ← 李园长（审批者）
├── journeys/
|   └── curriculum-director-monthly-proposal.md ← 6 阶段旅程 + 8 个机会点
└── processes/
    ├── monthly-proposal-creation.md            ← 主流程（5 决策点、4 角色）
    └── activity-design.md                      ← 活动设计子流程（嵌套循环）
```

---

4. 阶段二：Domain Model（领域建模）

4.1 Step 1: Event Clusters → Domains

```
Write studio/changes/course-workshop/domain-canvas.md
  → File created successfully
```

7 个域分类：

Domain	Events	Classification
主题分析	E1–E3	Supporting
标准制定	E4–E6	Supporting
问题与结构设计	E7–E11	**Core**
活动设计	E12–E15	**Core**
资源管理	E16–E18	Supporting
内容输出	E19–E20	Generic（不建插件）
质量保障	E21–E25	**Core**

域间关系图：

```
[主题分析] ──→ [标准制定] ──→ [问题与结构设计]
                                  |
                                  ▼
                            [活动设计] ──→ [资源管理]
                                  |              |
                                  ▼              ▼
                            [内容输出] ←──────────┘
                                  |
```

```
                          ▼
                      [质量保障]
```

4.2 Step 3: Behavior Matrix

```
Write studio/changes/course-workshop/behavior-matrix.md
  → File created successfully
```

6 个 Gap 识别：

Gap	Opportunity
无课标知识库	内置课标索引 + 自动匹配
无4C标准参照表	内置年龄段×4C能力矩阵
无驱动问题质量标准	开放性/可探究性自动评分
无活动时长估算模型	基于步骤数和年龄的时长预估
无资源数据库	PBL Box 物料数据库 + 自动匹配
无网络图生成工具	结构化网络图自动生成

4.3 Step 5: Opportunity Assessment

```
Write studio/changes/course-workshop/opportunity-brief.md
  → File created successfully
```

Impact × Feasibility 评分：

Plugin	Impact	Feasibility	Score	Priority
workshop-designer	5	4	**20**	2
workshop-quality	4	5	**20**	4
workshop-insight	4	4	**16**	3
workshop-resource	5	3	**15**	5
workshop-core	3	5	**15**	1 (基础设施)

Build Roadmap：

```
Phase 1 (MVP):   workshop-core + workshop-designer
Phase 2:         workshop-insight + workshop-quality
Phase 3:         workshop-resource + network-map enhancement
```

4.4 Step 6: Domain Map + Plugin Workspaces

```
Write studio/changes/course-workshop/domain-map.md
```

```
  → File created successfully
```

确定的 5 个插件：

Plugin	Role	Description	Dependencies
workshop-core	core	工作区初始化、状态管理、预案归档	—
workshop-designer	core	驱动问题→网络图→线索→活动→预案	workshop-core
workshop-insight	add-on	主题分析、先前经验、4C映射	—
workshop-quality	add-on	课标检查、预案审核	—
workshop-resource	add-on	资源匹配、资源校验	workshop-core

Collection Pattern: Core + Add-ons

```
workshop-core (zero deps)
    ↑
    ├── workshop-designer (depends: core)
    └── workshop-resource (depends: core)

workshop-insight  (zero deps)
workshop-quality  (zero deps)
```

创建 Plugin Workspaces：

```
mkdir studio/changes/workshop-core/
mkdir studio/changes/workshop-designer/
mkdir studio/changes/workshop-insight/
mkdir studio/changes/workshop-quality/
mkdir studio/changes/workshop-resource/

Write studio/changes/workshop-core/status.json       → { type: "plugin", phase: "planning", skills: {} }
Write studio/changes/workshop-designer/status.json    → { type: "plugin", phase: "planning", skills: {} }
Write studio/changes/workshop-insight/status.json     → { type: "plugin", phase: "planning", skills: {} }
Write studio/changes/workshop-quality/status.json     → { type: "plugin", phase: "planning", skills: {} }
Write studio/changes/workshop-resource/status.json    → { type: "plugin", phase: "planning", skills: {} }

Write studio/changes/course-workshop/status.json      → plugins: ["workshop-core", "workshop-designer", "workshop-insight", "workshop-quality", "workshop-resource"]
```

4.5 阶段二新增产出

```
studio/changes/course-workshop/
├── domain-canvas.md          ← 7 个域分类
├── behavior-matrix.md        ← Actor×Action×Event×Data 矩阵 + 6 个 Gap
├── opportunity-brief.md      ← Impact×Feasibility 评分 + 路线图
```

```
├── domain-map.md              ← 插件架构 + 依赖图

studio/changes/workshop-core/status.json
studio/changes/workshop-designer/status.json
studio/changes/workshop-insight/status.json
studio/changes/workshop-quality/status.json
studio/changes/workshop-resource/status.json
```

---

5. 阶段三：Skill Design（技能设计）

5.1 workshop-designer (5 skills)

```
Write studio/changes/workshop-designer/skill-map.md
  → File created successfully
```

Skills:

Skill	Description	Inputs	Outputs	Complexity
driving-question	驱动性问题生成+开放性验证	月度主题, 年龄段	driving-question.md	Simple+Agent
network-map	主题网络图构建	driving-question.md	network-map.md	Simple
inquiry-scaffold	探究线索拆分(3条递进)	driving-question.md + network-map.md	inquiry-clues.md	Simple+Agent
activity-design	活动序列编排	inquiry-clues.md	activities/clue-{N}.md	Moderate+Agent
proposal-generate	预案文档组装(华美5段式)	所有前序产出	proposal.md	Moderate

Data Flow:

```
[driving-question] → [network-map] → [inquiry-scaffold] → [activity-design] ×3 →
[proposal-generate]
```

5.2 workshop-core (3 skills)

```
Write studio/changes/workshop-core/skill-map.md
  → File created successfully
```

Skill	Description	Complexity
init	工作区初始化	Simple
status	状态查看	Simple

Skill	Description	Complexity
promote	预案归档	Simple

5.3 workshop-insight (3 skills)

```
Write studio/changes/workshop-insight/skill-map.md
  → File created successfully
```

Skill	Description	Complexity
theme-analysis	主题教育价值分析	Simple+Agent
prior-knowledge	先前经验评估(认知/技能/情感)	Simple+Agent
competency-mapping	4C映射+学习目标生成	Simple+Agent

5.4 workshop-quality (2 skills)

```
Write studio/changes/workshop-quality/skill-map.md
  → File created successfully
```

Skill	Description	Complexity
standards-check	自动规则检查(8条)	Moderate+Agent
proposal-review	多专家评审(3专家并行)	Moderate+Agent

5.5 workshop-resource (2 skills)

```
Write studio/changes/workshop-resource/skill-map.md
  → File created successfully
```

Skill	Description	Complexity
resource-planner	资源匹配+分类	Simple
resource-check	完整性校验	Simple

5.6 更新所有 status.json（添加 skills）

```
Write workshop-designer/status.json → skills: { driving-question: "draft", network-map: "draft", ... }
Write workshop-core/status.json     → skills: { init: "draft", status: "draft", promote: "draft" }
Write workshop-insight/status.json  → skills: { theme-analysis: "draft", prior-knowledge: "draft", competency-mapping: "draft" }
Write workshop-quality/status.json  → skills: { standards-check: "draft", proposal-review: "draft" }
```

```
Write workshop-resource/status.json → skills: { resource-planner: "draft",
resource-check: "draft" }
```

6. 阶段四：Spec Generate（规格生成）

6.1 workshop-designer — briefs + manifests + SKILL.md skeletons + commands

```
Write studio/changes/workshop-designer/brief.md           → File created
Write studio/changes/workshop-designer/plugin.json.draft   → File created

mkdir -p studio/changes/workshop-designer/skills/{driving-question,network-
map,inquiry-scaffold,activity-design,proposal-generate}
mkdir -p studio/changes/workshop-designer/commands/

Write skills/driving-question/SKILL.md     → 46 lines (skeleton)
Write skills/network-map/SKILL.md          → 46 lines (skeleton)
Write skills/inquiry-scaffold/SKILL.md     → 45 lines (skeleton)
Write skills/activity-design/SKILL.md      → 46 lines (skeleton)
Write skills/proposal-generate/SKILL.md    → 45 lines (skeleton)

Write commands/driving-question.md         → File created
Write commands/network-map.md              → File created
Write commands/inquiry-scaffold.md         → File created
Write commands/activity-design.md          → File created
Write commands/proposal-generate.md        → File created
Write commands/design.md                   → 流水线命令（串联 5 个 skills）
```

6.2 workshop-core — via background Agent

```
Agent "Generate workshop-core specs" launched
  → 9 tool uses · 33s

Output:
  Write brief.md, plugin.json.draft
  Write skills/init/SKILL.md, skills/status/SKILL.md, skills/promote/SKILL.md
  Write commands/init.md, commands/status.md, commands/promote.md
  → 8 files created
```

6.3 workshop-insight — via background Agent

```
Agent "Generate workshop-insight specs" launched
  → 9 tool uses · 32s

Output:
  Write brief.md, plugin.json.draft
  Write skills/theme-analysis/SKILL.md, skills/prior-knowledge/SKILL.md,
skills/competency-mapping/SKILL.md
  Write commands/theme-analysis.md, commands/prior-knowledge.md,
```

```
commands/competency-mapping.md
  → 8 files created
```

6.4 workshop-quality + workshop-resource — via background Agent

```
Agent "Generate quality+resource specs" launched
  → 13 tool uses · 54s

Output:
  workshop-quality: brief.md, plugin.json.draft, skills/standards-check/SKILL.md,
skills/proposal-review/SKILL.md, commands/standards-check.md, commands/proposal-
review.md
  workshop-resource: brief.md, plugin.json.draft, skills/resource-planner/SKILL.md,
skills/resource-check/SKILL.md, commands/resource-planner.md, commands/resource-
check.md
  → 12 files created
```

6.5 阶段四产出统计

```
=== Total files === 70

=== By plugin ===
course-workshop:  12 files
workshop-core:    10 files
workshop-designer: 15 files
workshop-insight: 10 files
workshop-quality:  8 files
workshop-resource: 8 files

agents: 3 files
```

6.6 Git Commit

```
git add studio/agents/ studio/changes/
git commit -m "feat: complete 4-phase planning pipeline for course-workshop-
plugins"
  → [main bb54fe1] 66 files changed, 5307 insertions(+)
```

---

7. SKILL.md 充实（从骨架到完整指令）

7.1 workshop-designer（5 个 skills，手动充实）

driving-question (46 → 245 lines)

```
Write skills/driving-question/SKILL.md
  → File updated successfully
```

新增内容： – Expert Discovery（动态专家加载） – Pre-check（工作区检测 + 可选 enrichment 文件读取） – Step 1: 引导式上下文收集（三要素引导 prompt） – Step 2: 生成 5 个候选问题 + 6 条设计规则表 + 年龄语言指南 – Step 3: 4 维度评分矩阵（开放性/可探究性/年龄适切/多维度，each 1–5）+ 详细 rubric – Step 4: Expert Review（课程专家自动审核 + 常见修正模式） – Step 5: 结构化呈现（推荐问题 + 评分 + 备选 + 确认交互） – Step 6: 输出格式模板 + status.json 更新

network-map (46 → 166 lines)

```
Write skills/network-map/SKILL.md
  → File updated successfully
```

新增内容： – Pre-check（读取 driving-question.md） – Step 1–2: 从探究维度生成子问题（2–4 per dimension） – Step 3: 叶节点生成（2–4 per sub-question，action-oriented） – Step 4: 五大领域覆盖标注（≥ 3/5 必须覆盖） – Step 5: 文本 mind map 图格式 – Step 6–7: 验证 + 输出模板

inquiry-scaffold (45 → 174 lines)

```
Write skills/inquiry-scaffold/SKILL.md
  → File updated successfully
```

新增内容： – Expert Discovery（课程专家 + 心理学家双角色） – Step 1: 网络图分析（concrete → abstract 光谱排序） – Step 2: 3 条线索设计（具象体验 → 问题解决 → 整合展示）+ 设计规则 – Step 3: 4C 分配（年龄×4C 定义矩阵 + 常见映射错误表） – Step 4: 学习目标（年龄×动词指南） – Step 5: 关键词提取 – Step 6: 双专家并行审核 – Step 7–8: 呈现 + 输出模板

activity-design (46 → 220 lines)

```
Write skills/activity-design/SKILL.md
  → File updated successfully
```

新增内容： – Expert Discovery（教学设计师 + 心理学家） – Step 1: 活动数量规划 – Step 2: 活动设计（编码规范 PBL-Cx-y + 5 种活动类型步骤模板 + Tips 编写规则 + 资源标注规则） – Step 3: 时长检查（步骤数×分钟估算公式，按年龄段） – Step 4: 序列检查（逻辑递进 + 类型多样性） – Step 5: 目标覆盖检查 – Step 6: 双专家审核 – Step 7–8: 验证 + 输出模板

proposal-generate (45 → 135 lines)

```
Write skills/proposal-generate/SKILL.md
  → File updated successfully
```

新增内容： – Pre-check（必需/可选输入文件清单） – Step 1–5: 华美 5 段式逐段组装指令 – 01 项目概览: theme-analysis.md or auto-generate – 02 项目标准: competency-mapping.md + prior-knowledge.md or extract from inquiry-clues – 03 项目启动: driving-question + network-map – 04 项目探究: inquiry-clues + activities (bilingual activity table) – 05 项目展示: auto-generate from Clue 3 – Step 6-7: 格式化 + 输出

7.2 workshop-core（3 个 skills，via background Agent）

```
Agent "Flesh out workshop-core skills" launched
  → 9 tool uses · 65s

Results:
  init/SKILL.md:    92 lines (was skeleton)
    → Pre-check, inline config.yaml, directory structure, summary + next steps
  status/SKILL.md:  74 lines
    → Scan changes + archives, domain vs plugin distinction, dashboard table, next-action suggestions
  promote/SKILL.md: 93 lines
    → Phase validation, production structure build, archive with timestamp, domain workspace update
```

7.3 workshop-insight（3 个 skills，via background Agent）

```
Agent "Flesh out workshop-insight skills" launched
  → 10 tool uses · 283s

Results:
  theme-analysis/SKILL.md:      317 lines
    → Expert discovery, bilingual narrative rules, 5-domain coverage, curriculum standard entries, expert review
  prior-knowledge/SKILL.md:     354 lines
    → 3-dimensional assessment (cognitive/skills/emotional), age-band differentiation, entry format rules, 8 validation checks
  competency-mapping/SKILL.md:  373 lines
    → 4C behavioral manifestation table, 6 common misclassification errors, age-appropriate verb guide, 10 validation checks
```

7.4 workshop-quality + workshop-resource（4 个 skills，via background Agent）

```
Agent "Flesh out quality+resource skills" launched
  → 18 tool uses · 295s

Results:
  standards-check/SKILL.md:   393 lines
    → 8 quality rules with procedures: keyword scanning, age-verb matrix, 4C content indicators, step-count duration formula, progression keywords, regex code extraction, per-activity resource count, CJK/Latin scanning
  proposal-review/SKILL.md:   364 lines
    → 3 parallel expert invocations, 9 sub-scores, consensus/conflict resolution, top 5 recommendations
  resource-planner/SKILL.md:  268 lines
    → 4 categories with heuristics, quantity estimation formulas, specificity rules, PBL Box order summary
```

```
  resource-check/SKILL.md:    267 lines
    → 5 validation checks, catalog lookup chain, duplicate detection, category
correctness rules
```

7.5 充实结果统计

```
=== SKILL.md Line Counts ===
  74 workshop-core/skills/status/SKILL.md
  92 workshop-core/skills/init/SKILL.md
  93 workshop-core/skills/promote/SKILL.md
 135 workshop-designer/skills/proposal-generate/SKILL.md
 166 workshop-designer/skills/network-map/SKILL.md
 174 workshop-designer/skills/inquiry-scaffold/SKILL.md
 220 workshop-designer/skills/activity-design/SKILL.md
 245 workshop-designer/skills/driving-question/SKILL.md
 267 workshop-resource/skills/resource-check/SKILL.md
 268 workshop-resource/skills/resource-planner/SKILL.md
 317 workshop-insight/skills/theme-analysis/SKILL.md
 354 workshop-insight/skills/prior-knowledge/SKILL.md
 364 workshop-quality/skills/proposal-review/SKILL.md
 373 workshop-insight/skills/competency-mapping/SKILL.md
 393 workshop-quality/skills/standards-check/SKILL.md

=== Total === 3,535 lines
```

---

8. References 编写（7 个领域知识文件）

8.1 workshop-designer references（via background Agent）

```
Agent "Write designer references" launched
  → 3 tool uses · 111s

Files created:
  references/pbl-methodology-guide.md   (194 lines)
    → PBL overview, 华美 5-step path, 三阶段九要素, driving question principles,
inquiry progression, activity best practices

  references/activity-coding-spec.md    (109 lines)
    → PBL-C{x}-{y} format, 4 rules, naming convention, activity table format, juice
shop worked example
```

8.2 workshop-insight + workshop-quality references（via background Agent）

```
Agent "Write insight+quality references" launched
  → 4 tool uses · 237s

Files created:
  workshop-insight/references/guidelines-3-6.md   (197 lines)
```

```
    → 5 developmental domains (Health/Language/Social/Science/Art), each with 2-3
goals + 3 age-band expectations

  workshop-insight/references/4c-framework.md      (193 lines)
    → 4C definition matrix, age-specific manifestations, observable indicators, 6
misclassification errors, verb guide

  workshop-quality/references/age-ability-matrix.md (140 lines)
    → Cognitive/motor/social/language/emotional abilities × 4 age bands, activity
feasibility matrix (15 activity types), time recommendations, 7 validation rules
```

8.3 workshop-resource references（via background Agent）

```
Agent "Write resource references" launched
  → 3 tool uses · 65s

Files created:
  references/pbl-box-catalog.md     (65 lines)
    → 4 categories (Art & Craft / Science / Print / Media), each item with CN/EN
name, quantity, notes

  references/resource-categories.md (130 lines)
    → 4 resource categories with definitions/characteristics/examples,
classification decision tree, specificity rules
```

8.4 References Git Commit

```
git add studio/changes/*/references/
git commit -m "docs: add 7 reference files for domain knowledge"
  → [main 1ee4ba0] 7 files changed, 1028 insertions(+)
```

---

9. Promote（推进到正式插件目录）

9.1 创建目录结构

```
mkdir -p .claude-plugin/
mkdir -p workshop-{core,designer,insight,quality,resource}/.claude-plugin
mkdir -p workshop-
{core,designer,insight,quality,resource}/{skills,commands,references,agents,templat
es}
```

9.2 写入 Marketplace Manifest

```
Write .claude-plugin/marketplace.json
  → 注册 5 个插件
```

```
Write .claude-plugin/plugin.json
  → marketplace 元数据
```

9.3 写入 Per-Plugin Manifests

```
Write workshop-core/.claude-plugin/plugin.json        → name, version, description,
keywords
Write workshop-designer/.claude-plugin/plugin.json     → + dependencies:
["workshop-core"]
Write workshop-insight/.claude-plugin/plugin.json      → no deps
Write workshop-quality/.claude-plugin/plugin.json      → no deps
Write workshop-resource/.claude-plugin/plugin.json     → + dependencies:
["workshop-core"]
```

9.4 复制文件到正式目录

```
# Skills, commands, references for each plugin
cp -r studio/changes/$plugin/skills/*     → $plugin/skills/
cp -r studio/changes/$plugin/commands/*   → $plugin/commands/
cp -r studio/changes/$plugin/references/* → $plugin/references/

# Agents to plugins that use expert discovery
cp studio/agents/*.md → workshop-designer/agents/
cp studio/agents/*.md → workshop-insight/agents/
cp studio/agents/*.md → workshop-quality/agents/

# Shared reference
cp workshop-insight/references/guidelines-3-6.md → workshop-quality/references/

# Config template
cp studio/config.yaml → workshop-core/templates/config.yaml
```

9.5 写入 CLAUDE.md

```
Write CLAUDE.md
  → 项目说明 + 结构 + 依赖 + 流水线 + 专家 + 测试命令 + references 索引
```

9.6 最终目录结构验证

```
--- workshop-core ---
.claude-plugin/plugin.json
commands/init.md, promote.md, status.md
skills/init/SKILL.md, promote/SKILL.md, status/SKILL.md
templates/config.yaml

--- workshop-designer ---
.claude-plugin/plugin.json
agents/child-development-psychologist.md, early-childhood-curriculum-expert.md,
instructional-designer.md
```

```
commands/activity-design.md, design.md, driving-question.md, inquiry-scaffold.md,
network-map.md, proposal-generate.md
references/activity-coding-spec.md, pbl-methodology-guide.md
skills/activity-design/SKILL.md, driving-question/SKILL.md, inquiry-
scaffold/SKILL.md, network-map/SKILL.md, proposal-generate/SKILL.md

--- workshop-insight ---
.claude-plugin/plugin.json
agents/child-development-psychologist.md, early-childhood-curriculum-expert.md,
instructional-designer.md
commands/competency-mapping.md, prior-knowledge.md, theme-analysis.md
references/4c-framework.md, guidelines-3-6.md
skills/competency-mapping/SKILL.md, prior-knowledge/SKILL.md, theme-
analysis/SKILL.md

--- workshop-quality ---
.claude-plugin/plugin.json
agents/child-development-psychologist.md, early-childhood-curriculum-expert.md,
instructional-designer.md
commands/proposal-review.md, standards-check.md
references/age-ability-matrix.md, guidelines-3-6.md
skills/proposal-review/SKILL.md, standards-check/SKILL.md

--- workshop-resource ---
.claude-plugin/plugin.json
commands/resource-check.md, resource-planner.md
references/pbl-box-catalog.md, resource-categories.md
skills/resource-check/SKILL.md, resource-planner/SKILL.md

=== Total plugin files === 54
```

9.7 Promote Git Commit

```
git add .claude-plugin/ workshop-core/ workshop-designer/ workshop-insight/
workshop-quality/ workshop-resource/ CLAUDE.md
git commit -m "feat: promote plugins to production directory structure"
  → [main 9dd76f6] 57 files changed, 5590 insertions(+)
```

---

10. 总量统计

Git History

```
14dbb84  Initialize studio workspace for plugin development
bb54fe1  feat: complete 4-phase planning pipeline for course-workshop-plugins
1ee4ba0  docs: add 7 reference files for domain knowledge
9dd76f6  feat: promote plugins to production directory structure
```

File Counts

Category	Count	Lines
SKILL.md files	15	3,535
Reference files	7	1,028
Command files	16	~160
Agent files	3 (×3 copies = 9)	~450
Plugin manifests	7	~85
Design artifacts (studio/)	12	~1,200
Skill maps	5	~350
Briefs + drafts	10	~200
Other (CLAUDE.md, config, status)	~15	~150
Total	**~100 unique files**	**~7,000+ lines**

Pipeline Phases Summary

Phase	Technique	Key Output	Files
0. 输入分析	PDF/Excel 逆向拆解	预案结构 + 方法论框架	0 (analysis only)
1. Event Storm	5 角色头脑风暴	25 事件 + 3 Personas + 1 Journey + 2 Processes + 8 Hotspots	7
2. Domain Model	域划分 + Canvas + Matrix + Opportunity	7 域 → 5 插件 + 依赖图 + 路线图	5
3. Skill Design	技能拆分 + 接口定义 + 数据流	15 skills 设计 + 实现顺序	5
4. Spec Generate	骨架生成 + 命令文件	15 SKILL.md + 16 commands + 5 briefs + 5 manifests	46
5. SKILL.md 充实	逐技能详细编写	3,535 行完整指令	15 (覆盖写入)
6. References	领域知识编写	7 个参考文档 (1,028 lines)	7
7. Promote	目录重组 + manifests	正式插件结构 (54 files)	57

实测命令

```
cd /Users/liuyameng/Codes/Plugins/courses-workshop-plugins

claude --plugin-dir ./workshop-core \
       --plugin-dir ./workshop-designer \
       --plugin-dir ./workshop-insight \
       --plugin-dir ./workshop-quality \
       --plugin-dir ./workshop-resource
```

```
# 然后输入:
/workshop-designer:design 我周围的人
```